MIZORAM PUBLIC SERVICE COMMISSION

TECHNICAL COMPETITIVE EXAMINATION FOR JUNIOR GRADE OF MIZORAM HEALTH SERVICES (COMBINED) (DENTAL SURGEON SUB-CADRE) UNDER HEALTH & FAMILY WELFARE DEPARTMENT, GOVERNMENT OF MIZORAM. (SEPTEMBER-2025)

TECHNICAL SUBJECT PAPER-II (DENTAL SURGEON)

Time Allowed : 3 hours FM : 200

All questions carry equal marks of 2 each.
Attempt all questions.

1. The following material is mistaken for caries on a radiograph
(a) Calcium hydroxide (b) Zinc oxide eugenol
(c) Zinc phosphate (d) Polycarboxylates

2. The main purpose of sealing the caries in deep cavities is
(a) To allow secondary dentine deposition before complete excavation
(b) To avoid removing the more tooth structure
(c) To kill the bacteria deep in the caries lesion
(d) All of the above

3. Principal component of the dental plaque is
(a) Dextrans (b) Polysaccharides
(c) Micro-organisms (d) Lactic acid

4. Caries activity is more in persons suffering with
(a) Xerostomia (b) Polysaccharides
(c) Parkinsonism (d) All of the above

5. The first priority in patient with multiple carious pulp exposures is
(a) Control of pain (b) Restoration of all the teeth
(c) Complete examination (d) Planning of treatment

6. Which of the following bacteria is primarily responsible for development of dental caries?
(a) Streptococcus mutans (b) Lactobacillus acidophilus
(c) Actinomyces viscosus (d) Streptococcus sanguis

7. What is the effect of xylitol on dental caries?
(a) Increases the risk of caries (b) Has no effect on caries
(c) Reduces the risk of caries (d) Reverses existing caries

8. A proximal carious lesion in a radiograph is
(a) Usually larger than clinically (b) Same as clinically
(c) Smaller than clinically (d) None of the above

9. Senile caries most frequently occurs on
(a) Proximal surface of the posterior teeth
(b) Proximal surface of the anterior teeth
(c) Root surfaces of both anterior and posterior teeth
(d) Proximal surfaces of both anterior and posterior teeth

10. The tooth designated as 36 is
(a) Left upper first molar
(b) Left lower first molar
(c) Right lower first molar
(d) Right upper first molar

11. What is the primary active ingredient in most dental bleaching products?
(a) Hydrogen peroxide
(b) Sodium perborate
(c) Carbamide peroxide
(d) Chlorine dioxide

12. What is the purpose of using a desensitizing toothpaste after dental bleaching?
(a) To enchance bleaching effect
(b) To prevent tooth decay
(c) To reduce tooth sensitivity
(d) To whiten teeth further

13. What is the concentration of hydrogen peroxide typically used in in-office dental bleaching?
(a) 10-15%
(b) 25-40%
(c) 15-25%
(d) 40-50%

14. Which of the following restorative material is known for its strength and durability?
(a) Amalgam
(b) Ceramic
(c) Composite resin
(d) Gold

15. Occlusal dovetail in class II inlay preparation
(a) Resist proximal displacement
(b) Increase resistance from occlusal forces
(c) Resists occlusal displacement
(d) All of the above

16. Localized necrosis of the pulp from cutting burs is
(a) Due to pressure
(b) Due to vibrations
(c) Due to heat
(d) All of the above

17. Filler used in composites is
(a) Quartz
(b) Aluminium
(c) Colloidal silica
(d) Glass

18. Composites used as dental sealant are composed of
(a) Glass ionomer
(b) Bis-GMA
(c) Methylmethacrylate
(d) EBA resins

19. Exposed dentin should be covered with_____when composites are used with acid etch technique
(a) Varnish
(b) Phosphoric acid
(c) Calcium hydroxide
(d) ZnOE cements

20. The best gingival margin for porcelain jacket crown
(a) Chamfer
(b) Knife edge
(c) Shoulder
(d) None of the above

21. Tooth discolouration results from
(a) Hyperaemia of pulp
(b) Pulpal abscess
(c) Pulp necrosis
(d) All of the above

22. A patient under fixed orthodontic therapy got his anterior tooth injured and became non-vital. You would like to
(a) Perform endodontic therapy without disturbing bands and arch wires
(b) Remove arch wire and perform endodontic therapy
(c) Extract the tooth as orthodontic tooth movement is not possible in non-vital tooth
(d) Prescribe antibiotics until orthodontic treatment is complete

23. Referred pain to ear is mainly from
(a) Mandibular molars same side
(b) Maxillary molars same side
(c) Mandibular anterior teeth same side
(d) Maxillary anterior teeth same side

24. Most common cause of endodontic failures in maxillary first molar is
(a) Missed fourth canal
(b) Periodontal involvement of pulpal lesion
(c) Vertical root fracture
(d) None of the above

25. Anaesthetic test is used
(a) To test the vitality of pulp
(b) routinely with other diagnostic tests
(c) To differentiate and determine the source of pain
(d) To anaesthesize tooth before endodontic procedure

26. First line of treatment in the case of an acute apical abscess with swelling is
(a) Establish drainage
(b) Prescribe analgesics
(c) Prescribe antibiotics
(d) No treatment reassurance

27. Disadvantages of using calcium hydroxide in pulpotomy of deciduous tooth-
(a) Internal resorption
(b) P.A. abscess
(c) External resorption
(d) Calcification of canals

28. The success of calcium hydroxide pulpotomy is determined by-
(a) Formation of calcification bridge
(b) Formation of hard base to support the restoration
(c) Continuation of root formation & apexogenesis
(d) All of the above

29. Solvent used along with hedgestroem file to remove gutta-percha.
(a) Alcohol 10%
(b) Chloroform
(c) sodium hypochlorite 5%
(d) formocresol

30. The filling of choice for primary root canals.
(a) ZnO-Eugenol
(b) Gutta-percha
(c) alcium hydroxide
(d) chloropercha

31. The root canal of mandibular 1st molar with greatest curvature is:
(a) Mesiobuccal canal
(b) Distobuccal canal
(c) Mesiolingual canal
(d) Distolingual canal

32. Recapitulation is-
(a) Using successively larger files to flare the canal
(b) Removing the debris with smaller instruments than the instrument that goes to apex
(c) Circumferential filing with H-file
(d) Using various types of files and reamers to enlarge canal

33. Which of the following technologies is used for 3D visualization of root canal anatomy?
(a) CBCT
(b) Computed tomography
(c) MRI
(d) Ultrasound

34. Which of the following weakens the endodontically treated tooth to a great extent?
(a) Access cavity preparation
(b) Dehydration of dentinal tubules
(c) Canal debridement and shaping
(d) Canal obliteration

35. The most desirable tissue reaction at apex after pulpectomy-
(a) Cementum deposition at apical foramen
(b) Bone deposition at apical foramen
(c) Proliferation of periodontal ligament into apical foramen
(d) Proliferation of epithelium into apical foramen

36. Internal resorption is common in-
(a) Upper molar
(b) Lower central incisors
(c) Upper central incisors
(d) Upper canines

37. The painful response that subsides quickly when stimulus is removed is characteristics of:
(a) Normal pulp
(b) Irreversible pulpitis
(c) Reversible pulpitis
(d) Necrotic pulp

38. The term "incompetent lips" refers to:
(a) Lips against which tongue thrusts during swallowing
(b) Lips that contract during swallowing
(c) Lips which do not seal when relaxes with the mandible in its rest position
(d) Lips which have some dryness

39. Main growth site of mandible is:
(a) Condyle
(b) Angle
(c) Ramus
(d) Coronoid process

40. Anterior open bite associated with class I relation is generally due to-
(a) Thumb sucking
(b) Abnormal swallowing
(c) Mouth breathing
(d) None of the above

41. Class III and a pseudo class III malocclusion can be differentiated by-
(a) Degree of crossbite
(b) Molar relation and canine relationship
(c) The presence of forward shift of mandible in pseudo class III during closing
(d) None of the above

42. Angle of convexity
(a) SNA
(b) SNB
(c) N-A-Pog
(d) ANB

43. Tooth movement easily accomplished by a removable appliances-
(a) Tipping
(b) Torquing
(c) Bodily
(d) Expansion

44. PMA index is used to record the status of-
(a) Gingival disease
(b) Root caries
(c) Carious teeth
(d) Periodontal disease

45. The essential difference between a child's toothbrush and an adult's toothbrush is that a child's toothbrush has:
(a) Smaller diameter bristles
(b) A smaller head size
(c) Softer bristles
(d) An angled handle

46. When treating rampant dental caries an initial step should be to:
(a) Apply topical fluorides to all teeth once a week
(b) Perform prophylaxis, instruct oral hygiene maintenance and continue to evaluate for improved habits
(c) Remove all softened tooth structure and seal with zinc oxide eugenol
(d) Restore all teeth as quickly as possible with stainless steel crown

47. A patient presents with a flabby ridge. The best treatment plan would be to:
(a) Make the denture without thinking about the flabby ridge as it does not affect the denture
(b) Remove the tissue surgically
(c) Relieve the denture in excessive areas
(d) None of the above

48. A patient is observed with a sharp mandibular ridge, the impression technique to be used will be:
(a) Minimum pressure impression
(b) Selective pressure impression
(c) More pressure impression
(d) Makes no difference since impression technique have not been proved

49. The stability of a denture refers to:
(a) Resistance against vertical forces
(b) Resistance against horizontal forces
(c) Resistance to removal in the opposite direction
(d) All of the above

50. The main disadvantage of relining and rebasing the complete denture is
(a) Loss of retention
(b) Allergic reaction to reliner
(c) Reduction of stability
(d) Change in centric relation

51. The means by which one part of the RPD opposes the action of retainer in function is called-
(a) Retention
(b) Support
(c) Stabilisation
(d) Reciprocation

52. The sprue during casting should be located-
(a) At the thinnest portion of the wax up
(b) At the thickest portion
(c) Anywhere on the casting
(d) At the margin of the pattern

53. The pontic for the mandibular posterior region should be-
(a) Point contact
(b) Bullet
(c) Sanitary
(d) Saddle type

54. Disinfection of impression material used in the mouth is done by:
(a) Autoclaving
(b) Use of chemical like glutaraldehyde
(c) Heat sterilization
(d) None of the above

55. Which finish line is most preferred in fixed prosthodontics?
(a) Knife edge
(b) Chamfer
(c) Shoulder
(d) Feather edge

56. Bacteria free zone of dentinal caries is:
(a) Zone of decomposed dentin
(b) Zone of bacterial invasion
(c) Zone of decalcification
(d) Zone of dentinal sclerosis

57. Widely accepted theory of dental caries-
(a) Proteolytic theory
(b) Proteolytic chelation theory
(c) Acidogenic theory
(d) Autoimmune theory

58. Oral foci of miller's are seen in:
(a) Dental caries
(b) Lichen planus
(c) Herpes simplex
(d) Syphilis

59. Which tooth in the permanent dentition is the most susceptible to dental caries?
(a) Maxillary 1st premolar
(b) Maxillary 2nd molar
(c) Mandibular 1st molar
(d) Mandibular 2nd molar

60. Early invading bacteria in carious lesions are called ?
(a) Microcosm
(b) Pioneer bacteria
(c) Advancing bacteria
(d) Anaerobic bacteria

61. The enzyme glucosyl transferase secreted by Streptococcus -mutans synthesizes glucans from:
(a) Glucose
(b) Fructose
(c) Sucrose
(d) Lactose

62. Home bleach is also called as:
(a) Walking bleach
(b) Thermocatalytic bleach
(c) Power bleach
(d) All of the above

63. Dentist prescribed home applied bleaching technique uses:
(a) 35% Hydrogen Peroxide
(b) 10% Carbamide peroxide
(c) 18% Hydrochloric acid
(d) Sodium perborate

64. Discoloration of tooth in tetracycline therapy is because of formation of ?
(a) Calcium orthophosphate
(b) Dicalcium phosphate dehydrate
(c) Calcium oxide
(d) Tetracalcium phosphate

65. Phoenix abscess is :
(a) Reversible pulpitis
(b) Acute apical periodontitis
(c) Acute exacerbation of an existing chronic inflammation
(d) Chronic abscess

66. The root canal instrument most likely to break during use is a :
(a) File
(b) Broach
(c) Reamers
(d) Spreaders

67. Sequence of endodontic instrumentation :
(a) Broach-reamer-file
(b) Reamer-broach –file
(c) File-reamer-broach
(d) File-broach-reamer

68. The best instrument for cutting dentin is :
(a) Reamer
(b) File
(c) Broach
(d) Explorer

69. Best method to sterilize absorbent points is :
(a) Autoclaving
(b) Chemical sterilizer
(c) Glass bead/ salt sterilizer
(d) Passing in ethanol flame

70. A 15% solution of EDTA has a pH of :
(a) 5.5
(b) 7.3
(c) 7.9
(d) 07

71. Step down technique is :
(a) Preparation from apical to middle third
(b) Preparation from apical to coronal third
(c) Preparation from coronal to middle third
(d) Preparation from coronal to apical third

72. Gutta percha can effectively be sterilized by ?
(a) Hot salt sterilizer
(b) Chemical solutions
(c) Autoclaving
(d) Dry heat

73. Ultrasonic devices operate at :
(a) 25-30 KHz
(b) 35-40 KHz
(c) 2-3 KHz
(d) 5-10 KHz

74. Advantages of NiTi over k-files?
(a) Increased flexibility
(b) Low resiliency
(c) Resistance to breakage
(d) Less expensive

75. The shape of head of a Gates Glidden drill is :
(a) Flame shaped
(b) Cone shaped
(c) Ball shaped
(d) Diamond shaped

76. Most common cause of RCT failure:
(a) Incomplete removal of PA cyst
(b) Non- obturation of accessory canals
(c) Incomplete debridement with improper obturation
(d) Large size of PA pathology

77. Which of the following is used as best root end filling :
(a) MTA (b) Amalgam
(c) ZOE (d) Calcium hydroxide

78. Technique that is best suited for preparing fine and curved canals
(a) Step back (b) Step down
(c) Crown down (d) Modified double flare technique

79. In case you suggest RCT what is your choice of irrigant ?
(a) Hydrogen peroxide (b) Saline
(c) Chlorhexidine (d) Sodium hypochlorite

80. Which of the following cements is most kind to pulp ?
(a) Glass ionomer (b) Polycarboxylate
(c) Silicate (d) Resin cement

81. Comparison of proportions can be made by:
(a) 't' test (b) Chi- square test
(c) ANOVA (d) Correlation & regression

82. Vipeholm study is used in :
(a) Oral cancer (b) AIDS
(c) Caries (d) Periodontal disease

83. An extracted tooth is discarded in which bag ?
(a) Yellow (b) Red
(c) Black (d) Blue

84. The fluoride form that has to be freshly prepared before use is :
(a) NaF (b) SnF
(c) APF solution (d) APF gel

85. Cross –sectional studies are carried on:
(a) Same individuals but of different mid- sagittal sections
(b) Different individuals but of same age groups
(c) Same individuals at different ages
(d) Different individuals of different sagittal group

86. The Indian Government passed the " Cigarettes and other Tobacco Products Act" to prohibit advertisement of and regulate the production , supply and distribution of tobacco products in the year
(a) 1973 (b) 1983
(c) 1993 (d) 2003

87. Dentist act was introduced on:
(a) 27th March, 1948 (b) 29th March, 1948
(c) 27th March, 1949 (d) 29th March, 1949

88. Clinical features of mouth breathing is:
(a) Pigeon face appearance (b) Proclined mandibular anteriors
(c) Shallow and flat maxillary arch (d) Retroclined maxillary anteriors

89. Acromegaly is associated with:
(a) Class 1 malocclusion
(b) Class 1 cross bite
(c) Class 2 malocclusion
(d) Class 3 malocclusion

90. The term orthodontics was coined by:
(a) Lefelon of France
(b) Hunter
(c) Carabeli
(d) Norman Kingsley

91. New born infants are:
(a) Obligatory oral breathers
(b) Obligatory nasal breathers
(c) Predominantly oral breathers
(d) Nasal and oral breathers

92. Considering the growth of the face in all three planes, growth ceases last in which direction ?
(a) Antero posterior
(b) Sagittal
(c) Transverse
(d) Vertical

93. The average 'Leeway space' available in each half of the maxilla is approximately:
(a) 0.9 mm
(b) 2.9 mm
(c) 4.0 mm
(d) 6.9 mm

94. The unit of the FPD that attaches the abutment to the suspended unit is:
(a) Connector
(b) Abutment
(c) Pontic
(d) Retainer

95. The type of gold alloy used in fixed partial dentures is:
(a) Soft gold
(b) Hard gold
(c) Medium gold
(d) Extra hard gold

96. In FPD, retention is provided by:
(a) Cement
(b) Parallel walls
(c) Grooves
(d) Proximal slice

97. Dicor is:
(a) Castable ceramic
(b) Metavite
(c) Vitallium
(d) Vita ceramic

98. 'Lost Salt Technique' is used to fabricate:
(a) Maryland bridge
(b) Rochette bridge
(c) Cantilever bridge
(d) Virginia bridge

99. Sequence of steps in RPD:
(a) Guide plane, axial contouring, rest seat
(b) Axial contouring , guide plane , rest seat
(c) Axial contouring , rest seat , guide plane
(d) None of the above

100. The ideal occlusal clearance for a complete cast crown is:
(a) 0.5 m
(b) 1.5 mm
(c) 2 m
(d) 2.5 mm

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